CV

Elias Nikolaos Pergantis, MEng, MS, PhD

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Mechanical engineer and applied scientist with a PhD in thermal systems and controls with a focus on optimal and learning-based control and 7+ years developing forecasting, optimization, and machine-learning algorithms for residential HVAC, distributed energy resources, and grid-interactive buildings. Fluent in Python, MATLAB/Simulink, Modelica, and AWS.

EDUCATION

Purdue University 06/2021- 12/2024

- D-PhD in Mechanical Engineering (MS received August 2023)
  - *Supervisors: Prof. Davide Ziviani, Prof. Kevin J. Kircher*
  - Concentration: Advanced Controls & Thermal Systems. GPA: 3.97/4.00
  - Research Focus: Deployment of Model-Predictive-Control at scale in the US residential building stock.

University of Michigan at Ann Arbor 09/2020 - 05/2021

- D-PhD in Mechanical Engineering (4.0/4.0 GPA) – Transferred to Purdue University
  - *Supervisors: Prof. Pramod S. Reddy and Prof. Edgar Meyhofer*
  - Research Focus: Use of surface structuring for utilization in nanophotonics.

University of Bristol – United Kingdom 09/2015 - 09/2019

- MEng in Mechanical Engineering with First Class Honors (4-year joint degree)
  - Grade: 74/100, GPA 3.92/4.0 *(9th in a class of 95 students)*
  - Concentration: Computational Fluid Dynamics and Renewable Power Generation (Wind & PV).

PROFESSIONAL AND RESEARCH EXPERIENCE

Lead Engineer, Trane Technologies Residential R&D Group, 6200 Troup Hwy, Tyler, TX 75707 01/2025 - present

- Lead development of next-generation smart HVAC and thermostat control algorithms, from model identification through HIL validation to production embedded deployment.
- Areas of work include model-based control (MPC and adjacent methods), machine-learning surrogates, model identification from lab and field data, and code generation from Modelica and MATLAB/Simulink toolchains.
- Contributions to date include nine patent applications on smart HVAC, thermostat, and residential energy control.

Advanced Technology Intern, Lennox International, PD&R, 1600 Metrocrest Dr 05/2023 - 07/2023
Carrollton, TX 75006, Supervisors: S. Rajan, R. Goel, R. Brahme.

- Flammable refrigerant leakage mitigation testing to meet 2023 UL 60335-2-40 standard.
- Sensor integration and multi-processing.
- Hardware in-the-loop (HIL) testing.

Graduate Research Assistant, Center of High-Performance Buildings, Ray W. 06/2021 - 10/2024
Herrick Laboratories, Purdue University, US – **Supervisors: Prof. Davide Ziviani, Prof. Kevin J. Kircher.**

Research Projects

Thesis Title: ***Field Demonstration of Predictive Home Energy Management***

- Investigation of machine learning integration with advanced controls, namely model-predictive control (MPC), for optimization of a heat pump's operation in cold climate.
- Extensive experience in field testing, data acquisition and system troubleshooting.
- Collaborated with Carnegie Mellon University on a head-to-head reinforcement learning (RL) and MPC comparison.

Supplemental projects

Electrochemical Looping Heat-Pump (06/2021 – 12/2022)

- NO-Vapor compression, Electrochemical Looping Heat Pump (NOVEL HP), DE-EE0008673.

- Prototyping and experimental verification of a novel fuel cell-based heat pump cycle.

Thermodynamic Cycles for two-compartment Ultra Low Temperature Vaccine Refrigeration (10/2021 – 04/2022)

- Thermodynamic modeling of advanced vapor compression cycles (IHX, Economization, Cascade) in EES.
- Optimized vapor compression and air cycles identified and published.

Advanced Cycles and Applications for High Temperature Heat-Pumping (04/2022 – 09/2022)

- Thermodynamic modeling of three stage compression heat pumps and their benefits for heat delivery.
- Technoeconomic analysis of synergetic waste heat recovery applications of HTHPs.

Graduate Student Instructor, Heat Transfer (ME 335), University of Michigan 01/2021 - 05/2021

- 300-level course on different modes of heat transfer
- Organized MatLab computing and programming sections.
- Assisted students in comprehension of course material & graded HWs and Exams.

Graduate Research Assistant, NanoHeat Transport Research Group, Faculty of Mechanical, Engineering, University of Michigan, US – Directors: Profs. E. Meyhofer & P. S. Reddy 09/2020 - 05/2021

- Research into how surface structuring and 1D photonic crystals can be used for enhanced TE in the infrared.
- Lurie Nanofabrication Facility, Clean Room Training.

Freelance Mathematics and Physics Tutor, Athens, Greece 09/2019 - 09/2020

- Privately tutored senior high school students for their Panhellenic examinations.
- Aided undergraduate students in advanced computational modelling relating to Heat Transfer and CFD.

Lab & Research Assistant, Biofluids Research Group, Faculty of Mechanical Engineering, Bristol University, UK – Director: Dr. Alberto Gambaruto 09/2018 - 06/2019

- Created a swallowing simulation software using 360-CT scans from the Fujita Hospital University in Japan.
- Collaborated with lab researchers to connect my Langrangian fluid solver with a surface morpher (MIRTK) and with a Least Squares automatic segmentation software.

Engineering & Research Intern, DNV GL Energy Group, Bristol, United Kingdom 06/2018 - 08/2018

- Worked on verifying the wake deflection aspects of DNV GL's real-time dynamic wind farm controller LongSim.

Supervisor: Dr. Ervin Bossanyi, Senior Principal Researcher.

- Performed k-ε CFD simulations of air over a wind turbine in StarCCM+ using DNV GL's cluster HORNET.

SCHOLARSHIPS & FELLOWSHIPS & AWARDS

- **2nd Best Student Paper at the 8th International High Performance Building Conference** 07/2024
- **Recognition award Mechanical Engineering Faculty, Purdue University** 03/2023
- **Best Poster Presentation, IEA Heat Pump Conference 2023** 05/2023
- **Grant-in-Aid, ASHRAE** 09/2023 – 09/2024
- **Research Assistantship, Faculty of Mechanical Engineering, Purdue University** 06/2021 – 12/2024
- **Onassis Foundation Scholar** 09/2022 – 12/2024
- **Departmental Fellowship, College of Engineering, University of Michigan** 09/2020
- **Uete, Dr. Hiroyuki and Reinhart Fellowship, University of Michigan** 09/2020
- **British Universities Chess Championship** (BUCA), **Bronze Medal 2019.** 03/2019
- **Scholarship, University of Bristol Summer Research Funding for Outstanding Students.** 06/2018 - 08/2018
- **Amphibious Vehicle 2nd Prize – University of Bristol Design Manufacturing Contest.** 12/2017

SOFTWARE & TOOLS

- **ML & data science:** Python (pandas, NumPy, SciPy, scikit-learn, PyTorch, TensorFlow), Jupyter
- **Optimization & control:** MPC, reinforcement learning (model-based), CVXPY/Gurobi, MATLAB, Simulink, Modelica, MATLAB Coder
- **Data infrastructure & cloud:** SQL, InfluxDB, AWS, time-series pipelines
- **Building & energy simulation:** EnergyPlus, ResStock, EES, Modelica

- **CFD & engineering:** StarCCM+, OpenFOAM, ANSYS, COMSOL
- **Communication & DAQ:** Modbus, BACnet, LabView
- **Laboratory:** PIV, laser fluoroscopy, nanofabrication, 3D printing, thermophysical instrumentation

PATENTS

- *Purdue University* - Logic and Apparatus for Cost-Effective Heat Comfort Control for Air-Source Single-Speed Unitary Heat Pump Systems (Patent Appl.) 01/2024
- *Purdue University* - Smart Controls for Electrical Distribution Protection in Homes (Patent Appl.) 03/2024
- *Trane Technologies* - Patent application on a novel heat pump control algorithm (Patent Appl.) 02/2025
- *Trane Technologies* - Patent application on a novel heat pump control algorithm (Patent Appl.) 01/2026
- *Trane Technologies* - Patent application on a novel heat pump control algorithm (Patent Appl.) 01/2026
- *Trane Technologies* - Patent application on a novel thermostat control algorithm (Patent Appl.) 01/2026
- *Trane Technologies* - Patent application on a novel user-friendly energy saving feature (Patent Appl.) 01/2026
- *Trane Technologies* - Patent application on a novel HVAC control algorithm (Patent Appl.) 2026
- *Trane Technologies* - Patent application on a novel HVAC control algorithm (Patent Appl.) 2026
- *Trane Technologies* - Patent application on a novel HVAC control algorithm (Patent Appl.) 2026
- *Trane Technologies* - Patent application on a novel HVAC control algorithm (Patent Appl.) 2026

SOCIETIES & AFFILIATIONS

- Technical Chamber of Greece (Mechanical Engineering section) Lifetime membership
- International Building Performance Simulation Association (IBPSA) 06/2022 - present
- American Society of Heating, Refrigerating and Air-Conditioning Engineers 07/2021 - present
- Air Conditioning, Heating and Refrigeration Institute (AHRI) 07/2021 - present
- International Institute of Refrigeration (IIR) 07/2021 - present
- Bristol University Chess Society Committee Member and Treasurer 09/2015 - 06/2019
- Bristol University Jazz Orchestra, Alto Saxophone 09/2018 - 07/2019
- Scout and Scout Leader 06/2003 - 08/2018

CONFERENCES

- 15th REHVA HVAC World Congress (CLIMA 2025), Milan, Italy 4 – 06/06/2025
- 19th International Refrigeration and Air Conditioning Conference, West Lafayette, USA 15 - 18/07/2024
- 8th International High Performance Building Conference, West Lafayette, USA 15 - 18/07/2024
- ASHRAE 2024 Winter Conference, Chicago, USA 20 - 24/01/2024
- 26th International Congress of Refrigeration, Paris, France 21 - 25/08/2023
- 14th IEA Heat Pump Conference, Chicago, USA 15 - 18/05/2023
- 2022 Building Performance Analysis Conference and SimBuild, Chicago, USA 14 - 16/09/2022
- 19th International Refrigeration and Air Conditioning Conference, West Lafayette, USA 11 - 14/07/2022
- ASHRAE 2022 Winter Conference, Las Vegas, USA 29/01 - 02/02/2022
- 18th International Refrigeration and Air Conditioning Conference, West Lafayette, USA 24 - 28/06/2021
- 5th PRIMARE CONFERENCE, Cardiff University 2018 15 - 17/06/2021

REVIEWER (JOURNAL)

Peer reviewer — 30+ manuscripts across leading energy and buildings journals including *Applied Energy*, *Energy Policy*, *Energy Conversion and Management*, *Building and Environment*, *Energy and Buildings*, *Sustainable Energy Grids and Networks*, and *Science and Technology for the Built Environment*. Reviewer for 12+ ASHRAE, REHVA, and Purdue conferences (2022–present).

REVIEWER (CONFERENCES)

- ASHRAE 2026 Winter Conference, Las Vegas, USA 31/01 - 04/02/2026
- ASHRAE 2025 Annual Conference, Phoenix, USA 21 - 25/06/2025
- 15th REHVA HVAC World Congress (CLIMA 2025), Milan, Italy 04 – 06/06/2025
- ASHRAE 2025 Winter Conference, Orlando, USA 08 – 12/02/2025
- ASHRAE 2024 Annual Conference, Indianapolis, USA 22 - 26/06/2024
- ASHRAE 2024 Winter Conference, Chicago, USA 20 - 24/01/2024
- ASHRAE 2023 Annual Conference, Tampa, USA 24 - 28/06/2023

- 26th International Congress of Refrigeration, Paris, France 21 - 25/08/2023
- ASHRAE 2023 Winter Conference, Atlanta, USA 4 - 8/02/2023
- ASHRAE and SCANVAC, HVAC Cold Climate Conference 2023, Anchorage, USA 6 - 8/03/2023
- 19th International Refrigeration and Air Conditioning Conference, West Lafayette, USA 11 - 8/07/2022

SEMINARS

- Progress in Heat Pumps with Low GWP Refrigerants, IEA 2023 Chicago 05/15/2023
- Flexibility in energy grids provided by heat pumps, IEA 2023 Chicago 05/15/2023
- Investors' role in different parts of the value chain of heat pumps, IEA 2023 Chicago 05/15/2023
- Trane La Crosse branch (Commercial), Chiller factory tour 10/01/2022
- Updates on Flammable Refrigerants, 19th International Refrigeration and Air Conditioning Conference 07/10/2022
- Ejectors and their application, 19th International Refrigeration and Air Conditioning Conference 07/10/2022
- Building Technology and routes to Zero Net Emissions, ASHRAE 2022 Winter Conference 01/30/2022
- Combined Heat and Power (CHP) Technology and its future in the US, ASHRAE 2022 W. Conf. 01/31/2022

Sample publication list, please review my Google Scholar for an up-to-date list of journals, conference papers and preprints.

PUBLICATIONS - JOURNAL

- **Pergantis E. N.,** Priyadarshan, Altheeb N., Dhillon P., Ore, J., Ziviani D., Groll E. A., Kircher K. J. (2024). *Field demonstration of predictive heating control for an all- electric house in a cold climate, Applied Energy* 360 (2024) 122820.
- **Pergantis E. N.,** Dhillon P., Premer L., Lee, A. H., Ziviani D., Groll E. A., Kircher K. J. (2024). *Humidity-aware model predictive control for residential air conditioning: A field study. Building and Environment*, 266, 112093.
- Khabbazi, A. J., **Pergantis, E. N.**, Reyes Premer, L. D., Papageorgiou, P., Lee, A. H., Braun, J. E., Henze, G. P., Kircher, K. J. (2025). *Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review. Applied Energy, 399*, 126459
- **Pergantis E. N.,** Bani Issa, A. A. M., Ziviani D., Groll E. A., Kircher K. J (2024). The potential of smart load reduction for mitigating frost in domestic heat pumps. Journal in Preparation.
- Reyes Premer, L. D., **Pergantis, E. N.,** Semmelmann, L., Ziviani, D., Kircher, K. J. (2026). *Model predictive control lowers barriers to adoption of heat-pump water heaters: A field study. Energy Conversion and Management*, 348 (Part B), 120723.
- Zhu, M., **Pergantis, E. N.**, Kim, J., Dai, C., Nelson, J., Oh, J., Baby, A., Rodríguez-López, J., Groll, E. A., Braun, J., Ziviani, D. *A critical review of electrochemical heat pump technologies: Status, challenges, and perspectives. Applied Thermal Engineering*, 129232.
- **Pergantis E. N**, Reyes Premer, L. D., Lee, A. H., Priyadarshan, Liu, H., Groll, E. A., Ziviani, D., Kircher, K. J. (2025). *Protecting residential electrical panels and service through model predictive control: A field study. Applied Energy*, 386, 12552.
- Mulayim, O.B., **Pergantis, E.N.**, Reyes Premer, L.D., Chen, B., Qu, G., Kircher, K.J., Bergés, M. (2025). Comparative Field Deployment of Reinforcement Learning and Model Predictive Control for Residential HVAC. Under review, *Applied Energy*. arXiv:2510.01475.
- Lee, A.H., **Pergantis, E.N.**, Kircher, K.J. (2026). Data-driven estimation of design heating loads for HVAC equipment sizing. arXiv:2609.07619.

PUBLICATIONS - CONFERENCE

- **Pergantis, E. N.,** Bani Issa, A. M., Liang, C., Cai, J., Ziviani, D., Kircher, K. J. (2025). Frost accumulation reduction in air-source heat pumps using load regulation. 15th REHVA HVAC World Congress (CLIMA 2025), Milan, Italy (June 4–6, 2025).
- **Pergantis, E. N.,** Reyes Premer, L. D., Khabbazi, A. J., Priyadarshan, Wu, F., Ziviani, D., Kircher, K. J. (2025). Active current limiting control of residential appliances for breaker panel protection across the US: a parametric study, CISBAT 2025 (EPFL program listing; published in Journal of Physics: Conference Series 3140(5), 052027.
- Reyes Premer, L. D., Semmelmann, L., **Pergantis, E. N.,** Groll, E. A., Ziviani, D., Kircher, K. J. (2024). *Field Demonstration of Predictive Heat Pump Water Heater Control*, 8th International High Performance Buildings Conference at Purdue (West Lafayette, IN, USA).

- Khabbazi, A. J., **Pergantis, E.**, Reyes Premer, L., Kircher, K. J. (2024). What have we learned from field demonstrations of advanced commercial HVAC control? International High Performance Buildings Conference at Purdue (2024 Purdue Conferences / IHPBC), West Lafayette, IN, US.
- Sriram, L., Farha, A., Hoess, A. J., Ziviani, D., Groll, E. A., **Pergantis, E.N.**, Kircher, K. (2024). Development and Comparative Analysis of a Power-over-Ethernet (PoE) DC Lighting System for Residential Buildings, International High Performance Buildings Conference (Purdue e-Pubs proceedings)
- **Pergantis, E. N.**, Park, J., Priyadarshan, Bird, T. J., Ziviani, D., Kircher, K. J. (2024). Open-source tool-box for identifying envelope dynamics of detached residential buildings, 8th International High Performance Buildings Conference at Purdue (West Lafayette, IN, USA).
- **Pergantis, E. N.,** Reyes Premer, L. D., Lee, A. H., Priyadarshan, Liu, H., Groll, E. A., Ziviani, D., Kircher, K. J. (2024). Protecting residential electrical infrastructure through advanced control: The first field results, 8th International High Performance Buildings Conference at Purdue (West Lafayette, IN, USA).
- **Pergantis E. N.,** Sangamnerkar, A. S., Priyadarshan, Ore, J. P., Dhillon, P., Ziviani, D., Groll, E. A., Kircher, K. J. (2023). Sensors, storage, and algorithms for practical optimal controls in residential buildings, ASHRAE Annual Conference (Tampa, FL, USA; June 24–28, 2023). (Conference paper TA-23-C045).
- Farha, A., Ore, J., **Pergantis, E. N.**, Sriram, L., Ziviani, D., Groll, E. A. (2023). Design and Application of an in Situ IoT-Based Energy Monitoring System for Use in Real Time Building Modeling, ASHRAE Annual Conference (Tampa, FL, USA; in ASHRAE Transactions, Vol. 129, Part 2, paper TA-23-C052, p. 490).
- **Pergantis, E.N.**, Bani Issa, A. A. M., Brehm, J. K, Groll, E. A., Ziviani, D. (2022). Thermodynamic Analysis of the Cascade Economization Cycle for High Temperature Heat Pump Applications, 14th IEA Heat Pump Conference.
- **Pergantis, E.N.,** Braun, J. E., Groll, E. A., Ziviani, D., (2022). Investigation of Electrochemical Looping Heat Pump Technology in Heating Mode. Int'l Refrigeration and Air Conditioning Conf. Paper 2366.
- Bani Issa, A. A. M., **Pergantis, E. N.**, Brehm, J. K., Groll, E. A., Ziviani, D., (2022). Modeling of an Ultra-Low Temperature Refrigeration System for Independent Vaccines and Medical Supplies Storage. *Int'l Refrigeration and Air Conditioning Conf. Paper 2400.*
- Brehm, J. K, **Pergantis, E. N.**, Bani Issa, A. A. M., Groll, E. A., Ziviani, D. (2022). Thermodynamic Assessment of Air-Cycles for Ultra-Low-Temperature Refrigerated Container Applications. *Int'l Refrigeration and Air Conditioning Conf. Paper 2536.*
- **Pergantis, E. N.,** Gambaruto, A., *Developing a numerical swallowing simulation*, Research Archives, University of Bristol, September 2019 DOI: https://research-information.bris.ac.uk/en/studentTheses/developing-a-numerical-swallowing-simulation

LANGUAGES

- **English** (fluent), **French** (advanced), **Greek** (native), **Korean** (beginner).